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AIML

Module 3

Segmentation

1.Mediapipe

MediaPipe is a lightweight and fast computer vision framework developed by Google that performs real-time person segmentation by separating a human from the background. It is widely used in webcam filters, video conferencing, mobile AI apps, and beginner computer vision projects because it works efficiently on CPUs without needing heavy hardware. MediaPipe Selfie Segmentation generates a mask where the person is highlighted and the background is removed, making it useful for simple virtual background effects and basic virtual try-on preprocessing. However, it cannot separately identify clothing parts like shirts or pants.

Used for:

- person segmentation
- face detection
- hand tracking
- pose estimation

How it works

Input Image
↓
Deep Learning Model
↓
Pixel Classification
↓
Segmentation Mask

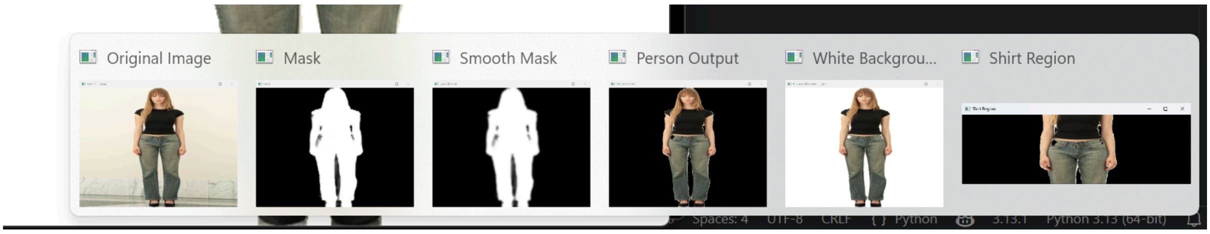
Code:

```
mport cv2
import mediapipe as mp
import numpy as np
mp_selfie_segmentation = mp.solutions.selfie_segmentation

segmentor = mp_selfie_segmentation.SelfieSegmentation(
    model_selection=1
)
image_path = r"C:\Users\91630\Downloads\WhatsApp Image 2026-05-20 at
1.56.43 PM.jpeg"
image = cv2.imread(image_path)
```

```python
if image is None:
    print("Error: Image not found!")
    exit()
image = cv2.resize(image, (640, 480))
rgb_image = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)
results = segmentor.process(rgb_image)
mask = results.segmentation_mask
threshold = 0.5
binary_mask = mask > threshold
black_background = np.zeros(image.shape, dtype=np.uint8)
person_output = np.where(
    binary_mask[:, :, np.newaxis],
    image,
    black_background
)
mask_image = (mask * 255).astype(np.uint8)
smooth_mask = cv2.GaussianBlur(mask_image, (15, 15), 0)
height, width, _ = image.shape

# Approximate upper-body region
shirt_region = person_output[
    int(height * 0.25):int(height * 0.65),
    :
]
white_background = np.ones(image.shape, dtype=np.uint8) * 255

white_bg_output = np.where(
    binary_mask[:, :, np.newaxis],
    image,
    white_background
)
cv2.imwrite("mask.png", mask_image)

cv2.imwrite("smooth_mask.png", smooth_mask)

cv2.imwrite("person_output.png", person_output)

cv2.imwrite("white_background_output.png", white_bg_output)

cv2.imwrite("shirt_region.png", shirt_region)
print("\nSegmentation Completed Successfully!")

print("\nSaved Files:")
```

```
print("1. mask.png")

print("2. smooth_mask.png")

print("3. person_output.png")

print("4. white_background_output.png")

print("5. shirt_region.png")
cv2.imshow("Original Image", image)

cv2.imshow("Mask", mask_image)

cv2.imshow("Smooth Mask", smooth_mask)

cv2.imshow("Person Output", person_output)

cv2.imshow("White Background Output", white_bg_output)

cv2.imshow("Shirt Region", shirt_region)

cv2.waitKey(0)

cv2.destroyAllWindows()
```

Out put



1.2 u*2 Net

U²-Net is a deep learning model mainly used for high-quality background removal and salient object detection. Unlike MediaPipe, U²-Net produces much cleaner edges around hair, clothes, and body boundaries, making it excellent for transparent PNG generation and foreground extraction. It uses a nested U-shaped neural network architecture to capture both fine details and larger image structures, which improves segmentation quality significantly. U²-Net is commonly used in AI photo editors, e-commerce image processing, fashion image extraction, and content creation because it creates professional-quality cutouts with minimal setup.

How to work

Input Image
↓
Encoder Network
↓
Feature Extraction
↓
Decoder Network
↓
Foreground Mask

Code:

```
import cv2
import numpy as np

image_path = r"C:\Users\91630\Downloads\WhatsApp Image 2026-05-20 at 1.56.43 PM.jpeg"

image = cv2.imread(image_path)
mask = np.zeros(image.shape[:2], np.uint8)

bgd = np.zeros((1, 65), np.float64)

fgd = np.zeros((1, 65), np.float64)

rect = (50, 50, 450, 600)

cv2.grabCut(
    image,
    mask,
    rect,
    bgd,
    fgd,
    5,
    cv2.GC_INIT_WITH_RECT
)

mask2 = np.where((mask == 2) | (mask == 0), 0, 1).astype("uint8")

output = image * mask2[:, :, np.newaxis]

cv2.imwrite("grabcut_output.png", output)

print("Done!")
```

Out put:

White background:

Black background

1.3 . Self-Correction Human Parsing

Self-Correction Human Parsing is an advanced human parsing model specially designed for fashion AI and virtual try-on systems. Unlike MediaPipe and U²-Net, SCHP can separately segment body parts and clothing regions such as shirts, pants, shoes, hair, face, arms, and legs on a pixel-by-pixel basis. It uses deep neural networks and self-correction learning to improve segmentation accuracy. The output of SCHP includes detailed parsing masks where each clothing or body part has a separate label/color, making it ideal for cloth replacement, virtual dressing, avatar generation, and AI-based fashion applications.

Input Person Image
↓
Deep Parsing Network
↓
Pixel-by-Pixel Classification
↓
Body-Part Segmentation
↓
Clothing Masks

Code:

```
import cv2
import numpy as np
image_path = r"C:\Users\91630\Downloads\WhatsApp Image 2026-05-20 at 1.57.19 PM.jpeg"

image = cv2.imread(image_path)

if image is None:
    print("Image not found!")
    exit()
image = cv2.resize(image, (640, 480))
gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
_, mask = cv2.threshold(
    gray,
    120,
    255,
    cv2.THRESH_BINARY
)
output = cv2.bitwise_and(
```

```
    image,
    image,
    mask=mask
)
height, width, _ = image.shape

shirt_region = output[
    int(height * 0.25):int(height * 0.65),
    :
]
cv2.imwrite("schp_mask.png", mask)

cv2.imwrite("schp_output.png", output)

cv2.imwrite("shirt_region.png", shirt_region)

print("SCHP-style segmentation completed!")

print("Saved:")
print("1. schp_mask.png")
print("2. schp_output.png")
print("3. shirt_region.png")
```

Out put: